ELECTROPHYSIOLOGICAL SIGNALS & SYSTEM ARCHITECTURE

A Comprehensive Technical Reference on Biosignal Capture, Muscle Tissue Mechanics, and Instrument Design

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1. Electrohysterography (EHG) & Uterine Monitoring

Electrohysterography (EHG) is a non-invasive method used to record the electrical activity of the uterine smooth muscle (myometrium) during pregnancy and labor. By placing surface electrodes on the maternal abdomen, clinicians can detect the weak microvolt electrical potentials generated by the uterus. Unlike traditional external tocodynamometry (TOCO), which evaluates mechanical abdominal displacement, EHG provides a direct, physiological assessment of uterine electrical activation.

The Two Core Wave Components of EHG

When raw EHG data is captured, it is structurally analyzed via signal processing into two distinct frequency domains:

- The Slow Wave (0.01 to 0.1 Hz): This ultra-low-frequency component forms the "mechanical envelope" of the contraction. It matches the macroeconomic physical timeline of a contraction, tracing the onset, peak amplitude, and relaxation phases.
- The Fast Wave (0.34 to 3.0 Hz): This higher-frequency component represents the underlying microscopic action potentials of individual myometrial cells firing concurrently. The fast wave is often subdivided into Fast Wave Low (FWL) and Fast Wave High (FWH). As labor approaches, the fast wave shifts toward higher frequencies and shows increased energy.

Signal Recognition & Feature Classification

Because the raw abdominal signal contains intense background noise (maternal ECG, fetal movement, respiratory interference), automated analysis relies on four core analytical domains:

1. **Amplitude (Time-Domain):** Measures such as Root Mean Square (RMS) evaluate total signal energy. RMS values spike markedly during coordinated contractions compared to baseline periods.
2. **Spectral (Frequency-Domain):** Median and Peak frequencies are calculated. As birth approaches, the formation of millions of new "gap junctions" improves cell synchronization, driving an upward shift in the signal's dominant frequencies.
3. **Time-Frequency Analysis:** Wavelet transforms are applied to track frequency shifts over time simultaneously, allowing precise filtering of maternal cardiac artifacts.

4. **Nonlinear Complexity:** Sample Entropy and Fuzzy Entropy track signal chaos. Uncoordinated pregnancy contractions exhibit high entropy (randomness), whereas synchronized labor contractions transition to low entropy (highly organized state).

2. Muscle Tissue Typologies and Electrical Behavior

The human body relies on three distinct variants of muscle tissue, each characterized by its own specialized cellular architecture, conduction speed, and electrophysiological presentation:

Muscle Type	Control & Structural State	Electrical Synchronization	Primary Biosignal Context
Smooth Muscle	Involuntary Non-striated, spindle cells	Slow, syncytial wave propagation via intercellular gap junctions.	EHG (Uterus) EGG (Stomach)
Cardiac Muscle	Involuntary Striated, branched cells	Rapid, highly synchronized, rhythmic polarization vectors.	ECG (Heart)
Skeletal Muscle	Voluntary Striated, linear fibers	Independent, extremely rapid firing of localized motor units.	EMG (Limbs / Torso)

3. Comparative Matrix of the "Electro-Grams"

Biopotential monitoring devices span several unique physiological layers across the body. The table below differentiates the most common diagnostic "electro-grams":

Signal Type	Full Name	Target Physiological Source	Tissue Type	Typical Bandwidth
EEG	Electroencephalogram	Cerebral Cortex (Neurons)	Central Nervous Tissue	0.5 – 100 Hz
ECG	Electrocardiogram	Myocardium (Heart)	Cardiac Muscle	0.05 – 150 Hz
EMG	Electromyogram	Skeletal Muscular Architecture	Skeletal Muscle	20 – 500 Hz
EHG	Electrohysterography	Myometrial Layer (Uterus)	Smooth Muscle	0.1 – 3.0 Hz
EOG	Electrooculogram	Corneo-Retinal Dipole (Eye)	Steady Electrical Dipole	DC – 30 Hz
EGG	Electrogastrogram	Gastric Smooth Muscle (Stomach)	Smooth Muscle	0.01 – 0.1 Hz
ERG	Electroretinogram	Retinal Photoreceptors & Cells	Neural / Retinal Tissue	0.1 – 1000 Hz

4. Hardware Bottlenecks & Instrument Interchangeability

The Fundamental Law of Biopotential Instrumentation

A primary biomedical engineering bottleneck prevents the arbitrary swapping of electrophysiological hardware: **The device's physical circuitry must perfectly match the unique voltage amplitude and frequency band of the target tissue.** Failure to align these properties causes signal destruction via amplifier saturation or hardware-level filter eradication.

The Two Core Engineering Bottlenecks:

- **The Amplitude Bottleneck (Amplifier Gain):** Brain waves (EEG) present at tiny amplitudes ($10 – 100\ \mu V$), requiring ultra-high gain amplification. Cardiac waves (ECG) are massive by comparison ($1 – 5\ mV$). Routing an ECG signal into an EEG amplifier causes instantaneous voltage clipping (saturation). Conversely, routing an EEG signal into an ECG system generates flatline readouts buried in noise.
- **The Speed Bottleneck (Frequency Bandpass Filtering):** Uterine contractions (EHG) occur at ultra-slow rates ($< 3\ ext\{\ Hz\}$), whereas active skeletal muscles (EMG) fire at high speeds ($ext\{up\ to\ \}\ 500\ ext\{\ Hz\}$). A biopotential instrument uses physical hardware filters to block ambient room noise. An EHG machine interprets high-speed EMG firing as "high-frequency noise" and wipes it out entirely. An EMG machine treats EHG signals as "low-frequency baseline drift" and filters it away.

5. Deep-Dive Instrument & Electrode Architecture

EEG (Electroencephalogram)

- **Biological Hurdle:** Brain signals must travel through cerebrospinal fluid, protective meninges, and an insulating bone barrier (the skull). Signals arrive at the scalp faint, spatially smeared, and weak.
- **Device Constraints:** Demands an ultra-high-gain configuration with an exceptionally high *Common Mode Rejection Ratio (CMRR)* to eliminate ambient alternating current room static.
- **Electrode Engineering:** Relies on **Silver/Silver Chloride (Ag/AgCl) cup electrodes** fixed inside a neoprene head-cap. Requires manual abrasion of the stratum corneum and the application of an electrolyte gel to bridge the high-resistance skull barrier.

ECG (Electrocardiogram)

- **Biological Hurdle:** The heart acts as a robust, synchronized electrical vector moving through highly conductive, fluid-filled chest tissues.
- **Device Constraints:** Moderate gain and broad bandpass parameters (***0.05 – 150 ext{ Hz}***) are required to concurrently record slow ventricular recovery patterns (T-waves) and sharp, sudden depolarization spikes (QRS complexes).
- **Electrode Engineering:** Employs large, **pre-gelled adhesive foam Ag/AgCl patches**. Electrodes are positioned far apart across the chest and extremities (forming *Einthoven's Triangle*) to map the physical vector of the heart in a three-dimensional field.

EMG (Electromyogram)

- **Biological Hurdle:** Highly localized, unsynchronized, high-speed firings from individual motor units in proximity.
- **Device Constraints:** Low-frequency hardware filters cut off everything under ***10 – 20 ext{ Hz}*** to omit bodily movement artifacts, while extending high-frequency capability past ***500 ext{ Hz}***.
- **Electrode Engineering:**
  Surface EMG: Closely-spaced dual metal bars positioned directly above the muscle center to prevent "crosstalk" from adjacent muscles.
  Intramuscular EMG: Ultra-fine, insulated **concentric needle electrodes** placed through the skin into the muscle belly to track single motor unit activations.

EHG (Electrohysterography)

- **Biological Hurdle:** The uterus is an extensive smooth muscle layer separated from the skin by amniotic fluid, a fetus, and maternal abdominal fat layers.
- **Device Constraints:** Uses steep low-pass hardware filters that exclude all frequencies above ***3 – 5 ext{ Hz}***, blinding the system to high-frequency motion artifacts.
- **Electrode Engineering:** Uses an expanded **multi-electrode surface array** configured across the maternal abdomen. Because smooth muscle waves spread slowly across a large area, wide spatial positioning is necessary to monitor the physical path and vector of the contraction wave.

EOG (Electrooculogram)

- **Biological Hurdle:** The eye functions as a constant, non-muscular electrical battery where the cornea is positive and the retina is negative. EOG measures the relative rotation of this steady electrical dipole.
- **Device Constraints:** Demands custom **DC-coupled amplifiers**. Standard AC-coupled biosignal units filter out unchanging baseline voltages. EOG systems must maintain these baseline DC measurements to determine gaze direction when an eye stops moving.
- **Electrode Engineering:** Highly stable, non-polarizable skin patches placed symmetrically around the orbital sockets to mitigate electrochemical voltage drift.

6. Alternative Modalities for Uterine & Fetal Health

- **Cardiotocography (CTG):** The standard clinical model utilizing external abdominal belts. It pairs an acoustic *Ultrasound Doppler* sensor (tracking fetal heartbeat) with a mechanical *Tocodynamometer (TOCO)* pressure sensor. TOCO records contraction timing via physical abdominal firmness, but cannot measure contraction force and is affected by maternal body mass index (BMI).
- **Intrauterine Pressure Catheter (IUPC):** The invasive clinical gold standard. A fluid-filled catheter is inserted directly into the amniotic sac via the cervix. It directly reads accurate fluid pressures in *ext{mmHg}*, but demands ruptured membranes and carries minor risks of infection.
- **Non-Invasive Fetal ECG (niFECG):** Uses the same maternal abdominal electrode configuration as EHG. Advanced internal algorithms filter out maternal signals to track clean, beat-to-beat fetal heart patterns without using ultrasound.
- **Magnetohysterography (MHG):** Tracks the magnetic fields generated by uterine electrical events using sensitive Superconducting Quantum Interference Device (SQUID) arrays. It delivers high-resolution spatial maps but requires an expensive, magnetically shielded environment.